

Combined Small-Angle Neutron Scattering, Diffusion NMR, and Molecular Dynamics Study of a Eutectogel: Illuminating the Dynamical Behavior of Glyceline Confined in Bacterial Cellulose Gels

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Chip J. Smith II, Durgesh V. Wagle, Nakara Bhawawet, Sascha Gehrke, Oldamur Hollóczy, Sai Venkatesh Pingali, Hugh O'Neill, and Gary A. Baker*

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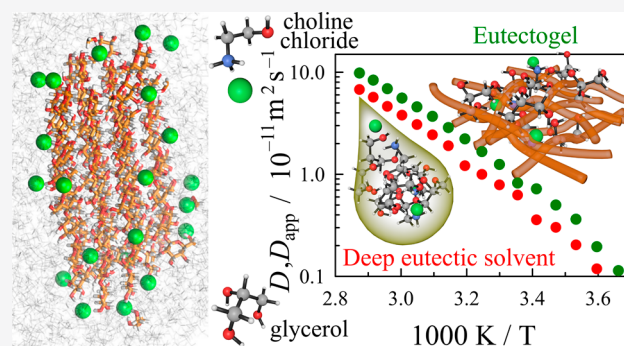
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ABSTRACT: A deep eutectic solvent (DES) entrapped in a bacterial cellulose (BC) network gives rise to a gelatin-like, self-supported material termed a bacterial cellulose eutectogel (BCEG). Although this novel material holds potential for numerous industrial, environmental, energy, or medical applications, little is known about the structural features or dynamical behavior within a eutectogel. In this work, we employ X-ray diffraction (XRD), nuclear magnetic resonance (NMR), and small-angle neutron scattering (SANS) to probe the structural and diffusive behavior of the prevailing DES glyceline (1:2 molar ratio of choline chloride:glycerol) confined within bacterial cellulose. XRD investigations demonstrate that the bacterial cellulose maintains its crystallinity even as the glyceline content approaches 95 wt % in the BCEG, an outcome corroborated by molecular dynamics (MD) simulations, which suggest minimal changes in the structural features of the cellulose chains due to the presence of glyceline. SANS measurements reveal a significant reduction in the radius of gyration (R_g) for BC in a BCEG compared to its hydrogel analogue, indicating a collapse in the microfibrillar structure that we attribute to removal of waters from the interfibrillar space due to a higher affinity of DES for water than for cellulose. Furthermore, SANS experiments suggest that the vast majority of DES is hosted within large micropores in the BCEG (i.e., mesoscopic confinement). Interestingly, proton NMR experiments disclose faster diffusional rates for choline and glycerol entrapped in a BCEG compared to neat glyceline. MD simulations offer the possible explanation that this diffusional acceleration results from significant migration of chloride from the bulk to cellulose microfibrillar surfaces, thereby reducing hydrogen bonding with choline and glycerol partners. This study provides the first comprehensive investigation into the structure and diffusional dynamics of glyceline within a eutectogel, offering insights into mass transport that should be useful for tailoring these novel materials to potential applications.



1. INTRODUCTION

Deep eutectic solvents (DESs) are new-generation solvents typically prepared by mixing a high-melting organic salt (hydrogen bond acceptor) with a neutral, organic molecule (which serves as a hydrogen bond donor) at a certain molar ratio to form a mixture that is a liquid at or near room temperature. For instance, when a quaternary ammonium salt such as choline chloride (302 °C melting point, m.p.) and urea (133 °C m.p.) are mixed in a 1:2 molar ratio, the resulting mixture (or DES) is a liquid near ambient temperature. The simple, cost-effective preparation of DESs from biodegradable, commodity materials minimizes the formation of chemical waste, making a strong case for DESs as environmentally friendly solvents. Since their inception, several different types of DESs have been synthesized from various combinations of ammonium salts (particularly, choline chloride) and hydrogen

bond donors (e.g., alcohols, amines, amides) allowing for the tailoring/designing of DESs toward a wide spectrum of applications, including electrolytes,¹ electrochemical metal deposition,² extractions,³ pharmaceuticals,⁴ and nanomaterials synthesis.^{2,5} Ionic liquids (ILs) and DESs are highly viscous solvents which can make their handling and utilization difficult. In order to overcome this obstacle, numerous approaches have been followed to confine or immobilize ILs and DESs in host

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materials such as silica or cellulose to render these solvents in a gel-like or solid-state format.^{1,6} Mesoscopic confinement of ILs and DESs is a highly practical approach as it not only facilitates handling but also affords retention of the fluid-like diffusive motion of molecules in the gel, an important consideration when applying these hybrids in electrolyte applications.^{7,8}

Deep eutectic gels (sometimes referred to as “eutectogels”) have become a recent focus in the ion gel literature with the emergence of a handful of published reports.^{1,8–16} In one early report of these gels, a resorcinol-formaldehyde gel was prepared via a polycondensation reaction inside aqueous and nonaqueous DESs for use in the preparation of carbon monoliths.¹⁰ The polycondensation reaction employing sodium carbonate and hydrochloric acid in a DES resulted in the unique ability to more evenly disperse carbon nanotubes inside these gels to result in increased conductivity. In this instance, the eutectogel was not actually the final product but was rather part of a sacrificial supramolecular template which was washed away to leave behind a skeleton for subsequent carbonization. It was not until 2016 that eutectogels were utilized directly, at which point multiple reports including those from the Prasad group began to surface on this topic.^{11,12} In their first report, a eutectogel was prepared by self-polymerization of 2-hydroxyethyl methacrylate (HEMA) in a DES comprising a 1:1.5 ratio of choline chloride:orcinol. Notably, HEMA was the only one of three tested monomers found to self-polymerize.¹¹ Gels prepared in this report were shown to display good flexibility, conductivity, and power density associated with values of 500% strain tolerance, >200 F g⁻¹, and 33 Wh kg⁻¹, respectively. The Panzer group later reported highly stretchable and compressible (up to 90%) gelatin-based eutectogels displaying excellent shape recovery.¹⁶ Gels containing 22 wt % gelatin and 88 wt % ethaline were investigated and achieved ionic conductivities comparable to hydrogels used in ionic skin devices, suggesting possible applications in this area. Using natural sources such as gelatin,¹⁶ polysaccharides,⁸ and amino acids¹⁵ as supporting matrices for eutectogels has the potential for additional “biologically friendly” innovations.

Although eutectogels represent a topic of growing interest, most reports to date focus on application or proof of concept, with relatively few reports focused on species mobility/diffusion within the support or investigating the effect of the DES component on the support matrix structure or dynamics. It is well established that the hydrogen-bonding network is essential for eutectic formation and strongly influences the local structure and dynamics within DESs;^{17–19} however, many details remain elusive. The rigorous understanding of the complexities of the hydrogen-bonding network is essential for the rational design of tailored and chemically diverse eutectogels. The detailed study of the dynamical structure within DESs and their binary mixtures with water, utilizing techniques such as small-angle neutron scattering (SANS), X-ray diffraction (XRD), nuclear magnetic resonance (NMR), and molecular dynamics (MD) simulations, offers invaluable insights into the underlying nanostructure, disorder, clustering (i.e., domain and transient cage formation), charge transfer, and phase behavior of the material.^{20–27}

While DESs have been thoroughly investigated to understand structure and behavior, very little work has been performed to understand eutectogel-confined DESs. Simulations based on XRD data illustrate that DESs have structures reminiscent to ILs, being locally heterogeneous, as shown

through the broad XRD peak (Q value between ~ 1.2 – $2 \text{ \AA}^{-1}$) measured in alkylamide/lithium perchlorate DESs.^{22,28} Most recently, neutron techniques including small-angle neutron scattering (SANS),²⁹ quasi-elastic neutron scattering (QENS),^{21,25} inelastic neutron scattering (INS),²³ neutron diffraction,²⁴ and neutron scattering^{20,21,26} have provided critical insight into the molecular-level behavior of DESs. Using neutron scattering, Faraone et al. demonstrated that the structure of glycine is controlled by strong hydrogen bonding, dominated by glycerol.²⁶ Neutron scattering experiments and QENS measurements have shown that the nanostructuring of malicine (1:1 choline chloride/malic acid) maintains 90% or more of its nascent (near-anhydrous) structure, even with large amounts ($\sim 42 \text{ wt } \%$) of water added.²⁰ QENS experiments revealed that localized diffusive motions are actually larger for choline than for glycerol in the DES glycine,²⁵ while NMR measurements show that glycerol diffuses faster in the long-range motions with diffusion coefficients of 3.8×10^{-12} for choline versus 5.2×10^{-12} for glycerol.³⁰ When water is added to the DES (up to 17.5 wt %), there are no major changes to the NMR diffusion measurements of glycine whereas changes are observed in the long-range diffusional rates for other DESs such as reline.³¹

In the current work, we introduce a new class of eutectogel inspired by our previous work with bacterial cellulose ionogels (BCIGs). Specifically, we incorporate the DES glycine within a bacterial cellulose host to generate a novel type of eutectogel we subsequently denote a glycine bacterial cellulose eutectogel (Gly-BCEG).^{32–34} The structure of the bacterial cellulose was examined through the methods of X-ray diffraction (XRD) and small-angle neutron scattering (SANS) to ascertain changes induced by glycine to the cellulose crystallinity and entanglement. Diffusion ordered spectroscopy (DOSY) was utilized to study the diffusive properties of glycine within the eutectogel. Complementary molecular dynamics simulations were employed to elucidate our experiments to afford a better understanding of the key differences in the structure and hydrogen bonding network for neat versus bacterial cellulose-confined glycine.

To the best of our knowledge, this work represents one of the first studies conducted on the diffusive behavior within a DES containing gel. Notably, this is also the first study aimed at understanding a eutectogel based upon bacterial cellulose which, it should be noted, has superlative mechanical features distinct from its plant-derived cellulosic counterparts.³⁵ Lastly, we suggest that BCEGs may hold benefits for the scientific community and industry as more sustainable alternatives in applications such as semisolid electrolytes, on-skin drug release films, separations and catalytic media, and sensory architectures, to name a few.

2. EXPERIMENTAL SECTION

2.1. Materials and Reagents. *Glucacetobacter xylinus* (strain ATCC 700178) was obtained from ATCC (Manassas, VA). Yeast extract, peptone, and dextrose were purchased from Fisher Scientific (Waltham, MA). Glycerol, choline chloride, sodium phosphate dibasic, and citric acid were acquired from Sigma-Aldrich (St. Louis, MO). Mannitol was purchased from Calbiochem Merck (Billerica, MA), and 200-proof (absolute) ethanol was obtained from Decon Laboratories (King of Prussia, PA).

2.2. Hydrogenated and Deuterated Glycine Synthesis. Conventional (hydrogenated) glycine (H-Gly) and

its deuterated counterpart (D-Gly) were prepared by combining the appropriate forms (i.e., hydrogenated or deuterated) of choline chloride and glycerol in a 1:2 molar ratio. The resulting mixtures were stirred and heated to 60 °C until a homogeneous fluid was obtained, followed by evacuation for 3 days to remove excess water.

2.3. Glycine Viscosity Measurements. Viscosities were measured using a Brookfield model DV-III+CP Rheometer with a CPE-42 Cone Spindle at 5 °C intervals over the temperature range from 5 to 80 °C.³⁶ Samples were dried under vacuum at 60 °C overnight prior to performing viscosity measurements.

2.4. Bacterial Cellulose Hydrogel and Alcogel Growth and Preparation. Bacterial cellulose (BC) was produced from cultures of *Gluconacetobacter xylinus* ATCC 700178, grown in a modified Hestrin-Schramm (H-S) medium following methods we reported previously.³² The H-S medium consists of 2% w/v mannitol, 0.5% w/v yeast extract, 0.5% w/v peptone, 0.25% w/v sodium phosphate dibasic, and 0.15% w/v citric acid. The medium was adjusted to pH 6 by addition of 6.0 M HCl and then autoclaved at 125 °C, followed by the addition of 1% v/v ethanol using a sterile 0.2 μm syringe filter. Starter cultures were prepared by adding 5% v/v of a mother medium to fresh medium in 50 mL polypropylene tubes (Fisher Scientific). After 3 days of starter culture growth, 5% v/v of starter culture was added to 250 mL of production cultures. The production cultures were allowed to grow for 4 days, the first pellicles were discarded, and then the second pellicles to develop were grown to the desired thicknesses, harvested with sterile forceps, and then washed in a 1% NaOH bath at 95 °C for 1 h. After the NaOH washing, pellicles were transferred to a deionized water bath (~500 mL) where they were rinsed, with water changes every 2 h until a neutral pH was registered. Alcogels were prepared from the cleaned hydrogels by soaking them in a 200-proof ethanol bath (100 mL) which was refreshed after 2 h. The preparation of alcogels and storage in ethanol allowed for maintenance of the cellulose pellicles indefinitely until required.

2.5. Bacterial Cellulose Eutectogel (BCEG) Preparation. BCEG samples for NMR and XRD measurements were prepared by soaking 2 cm × 2 cm × ~4 mm thick BC alcogels overnight in mixtures of glycine (~200 wt % relative to the starting alcogel). The vials were capped, vortexed to mix ethanol and glycine, and allowed to equilibrate overnight. The following day, the vials were uncapped and left open for ethanol evaporation for 2–7 days as needed, depending on atmospheric conditions. Following ethanol evaporation, the samples were weighed and their dimensions acquired and then placed under vacuum at 60 °C for 12 h, followed by reweighing. This drying procedure was continued for up to a week, until the gel mass remained constant from one measurement to the next. Each gel was then cut, using a razor blade, into thin slices the length of the gel. The resulting slices were packed into NMR or capillary tubes, removing air bubbles as necessary. SANS samples were prepared by cutting BC alcogels into ~1 mm thick rounds cut to a 13 mm diameter. These samples were then soaked in mixtures of either H₂O:D₂O (hydrogel) or D₂O:D-Gly (eutectogel) at various concentrations. The hydrogel sample containing a 99.6% mixture of mixed-isotope water was prepared from a mixture having a 22:78 H₂O/D₂O mass ratio. Eutectogel samples were prepared at a solvent loading of 99.6% from mixtures of either 99:1 or 50:50 mass ratios of D-Gly/D₂O. In

advance of preparing the above hydrogels and eutectogels, alcogels were first soaked in H₂O and D₂O baths, respectively. After soaking the cellulose in the first component of the mixture (i.e., H₂O or D₂O) the appropriate mass of the second component was added to prepare the desired gel. Upon addition of the second component, the vial was capped and sealed and the sample allowed to equilibrate for 1 week. To maintain the correct mass ratio of components, no subsequent drying was performed postequilibration.

2.6. Small-Angle Neutron Scattering (SANS). SANS measurements were performed with the CG-3 Bio-SANS instrument³⁷ at the High Flux Isotope Reactor (HFIR) facility of Oak Ridge National Laboratory. SANS measurements were performed using cylindrical Hellma cells with a 1 mm path length (model no. 120-QS 1.0 mm) and titanium cells with detachable quartz windows. A single instrumental configuration of the dual (main/wing) area detector system was employed to produce data spanning a Q-range from 0.003 to 0.8 Å⁻¹; wave vector Q is related to neutron wavelength (λ) and scattering angle (2θ) by $Q = (4\pi/\lambda) \sin \theta$. The main detector was located at 15.5 m from the sample position, and the curved wing detector is fixed at 1.13 m from the sample. The area detectors consist of 1 m × 1 m GE-Reuter Stokes Tubes for the main detector and 0.8 m × 1 m for the wing detector. The instrument resolution was defined using circular aperture diameters of 40 mm and 12 mm for the source and sample, respectively, and separated by a distance of 17.43 m. The relative wavelength spread Δλ/λ was set to 0.15. The scattering intensity profiles I(Q) versus Q were obtained by azimuthally averaging the processed 2D images, which were normalized to incident beam monitor counts, and corrected for detector dark current, pixel sensitivity, and scattering from the quartz cell.

The SANS data were fit to a combination of Guinier and power-law profiles as implemented in the multilevel Unified Fit model³⁸ applied in the IRENA³⁹ package of Igor Pro software by Wavemetrics. This model is best suited to extract structural details in hierarchical and disordered materials. Using a minimal set of fitting levels of the multilevel model, structural features such as particle shape, size, agglomeration pattern, and surface morphology can be extracted. The total SANS intensity profile is the sum of individual levels (i)

$$I(Q) = \sum_i I_i(Q)$$

$$= G_i e^{-Q^2 R_{gi}^2/3} + e^{Q^2 R_{gCO-1}^2/3} \times B_i \left\{ \frac{\left(\operatorname{erf} \left(\frac{QR_{gi}}{\sqrt{6}} \right) \right)^3}{Q} \right\}^{P_i}$$

$$+ C \quad (1)$$

where each level “i” contains an exponential and a power law part and adjacent levels are combined using an error function. R_{gi} is the radius of gyration of the particle of the *i*th structural level, P_i is the power-law exponent of the *i*th structural level, G_i is the scalar for the Guinier function of the *i*th structural level, and B_i is the scalar for the power-law function of the *i*th structural level. Additionally, R_{gCO-1} is the cutoff length scale of the power-law behavior of the *i*th structural level, set to the radius of gyration of the (*i* − 1)th structural level, and *C* is the Q-independent constant background intensity. For the

purposes of fitting in eq 1, the level-2 R_g values were fixed to a relatively large value, $R_g = 2000$ Å, since it is not possible to determine a value for level-2 R_g for either the Gly-BCEG gels or the control hydrogels because its size was larger than the accessible length scales (>1000 Å).

2.7. X-ray Diffraction (XRD). XRD measurements were performed on a Bruker Prospector instrument with an Apex II CCD detector and an IMuS microfocus Cu tube. XRD measurements were carried out on a BC aerogel, neat glycine, and BCEGs containing 18.4–94.8 wt % glycine. XRD was measured from $2\theta = 5$ – 45° with a 0.1° step size using a polyimide capillary sample holder. XRD diffractograms were fit to Voigt functions using PeakFit version 4.12. BC aerogel XRD data were first fit to determine parameters for subsequent BCEG fitting. Data were fit to five crystalline peaks plus an amorphous peak (Figure S1) following a previous study which reported using Voigt functions for all peaks to calculate the crystallinity index (CrI) instead of employing the peak height method, which can overestimate CrI.⁴⁰ The parameters were set with a 3% tolerance in the linear baseline fit, using a deconvolution width of 0.4, standard deviation (SD) setting, a filter setting of 71.2, with varying widths, and a 7.00 Amp%. CrI was calculated by dividing the integrated area of the BC crystalline peaks by the total area of the BC XRD peaks. The BC diffractogram fitting was also used to supply the parameters for the Scherrer equation (eq 2) in order to calculate crystallite size (τ).⁴¹

$$\tau = K\lambda/(\beta \cos \theta) \quad (2)$$

In eq 2, τ is the size perpendicular to the lattice plane represented by the peak in question, K is a constant that depends on the crystal shape (K is often close to unity), λ is the wavelength of the incident beam in the diffraction experiment, β is the full width at half-maximum (fwhm) in radians, and θ is the location of the peak.^{42,43} The d -spacing was calculated using the Bragg equation (eq 3), assuming a first order of diffraction.^{43,44}

$$d_{hkl} = \lambda/2 \sin \theta \quad (3)$$

Neat glycine diffractograms were well fit to three Voigt profiles (Figure S2). The first peak at the lowest 2θ ($\sim 5.7^\circ$) was present in all BC and BCEG samples, suggesting that it may arise as a background peak that was not completely removed during background subtraction. The broad, higher-angle peak is comparable to that seen previously for pyrrolidinium ILs and arises from intermolecular interactions within the DES.²⁸ This single DES peak at $2\theta \sim 19.4^\circ$ is well described by two broad, overlapping Voigt profiles.

XRD patterns for BCEG samples were fit by first importing the fitting parameters and peaks from the BC aerogel fit and then adding additional peaks, as appropriate, to account for the presence of glycine (Figures S3–S8). Each BC peak was locked at the 2θ position determined from the BC fit and glycine peak positions were allowed to float. Fitting of the BCEG diffractograms was iterated until an r^2 value of 0.9900 was attained. All locked peaks were then unlocked and fit iteration resumed until r^2 values of >0.9960 and F values of $>30\,000$ were achieved.

2.8. Nuclear Magnetic Resonance (NMR). No deuterium (No-D) NMR measurements^{45,46} were made on an Oxford AS600 NMR magnet with a Bruker AVIII HD 600 MHz console using a 5 mm CPTCI cryo-probe. ^1H NMR measurements were performed without spinning for both

eutectogels and the corresponding neat liquid. The spectra were then calibrated using an external deuterated 4,4-dimethyl-4-silapentane-1-sulfonic acid (DSS) in D_2O standard. The diffusion coefficients were measured with diffusion ordered spectroscopy (DOSY) NMR, using a gradient spin echo pulse sequence available through the TopSpin 3.5.7 Bruker software similar to that previously reported by Stejskal and Tanner.⁴⁷ Gradient strength was varied in 16 linear steps from 0.45–45.74 G cm^{-1} for the rectangular gradient pulse with smoothed edges, aiming to give a gradient integral of 90% of the square pulse, utilizing the SMSQ10.100 smoothing method. The stimulated echo height decays were fit using TopSpin software according to eq 4, where $I(g)$ and I_0 are the echo heights at gradient strengths g and 0, respectively, D is the diffusion coefficient, γ is the gyromagnetic ratio of ^1H , δ is the gradient-pulse length, and Δ is the diffusion time in the experiment. Self-diffusion within the BCEG (D_{app}) is an average of the dynamical processes occurring at the surface of the supporting matrix and the motions in the bulk-like regime. The δ (P30 in Bruker software) value was kept constant at 1 ms for all experiments. Values for Δ in all systems were optimized over the range of 50–100 μs .

$$I(g) = I_0 \exp\left[-D(2\pi\gamma\delta)^2\left(\Delta - \frac{\delta}{3}\right)1 \times 10^4\right] \quad (4)$$

2.9. Molecular Dynamics Simulations. Molecular dynamics (MD) simulations were carried out using LAMMPS⁴⁸ software. The molecular mechanical parameters were taken from the OPLS⁴⁹ force field. It should be noted that, in general, describing the dynamics of IL and DES systems is difficult through classical MD simulations. Applying polarizable force fields has been shown to improve the quality of the results;^{50–53} however, the computational demand of such simulations is significantly higher, while polarizable parameters are not yet established for DES systems. Indeed, polarizable models for DESs have not yet been tested extensively; therefore, the improvement they may bring to the modeling of these materials remains in question. Alternatively, charge scaling schemes in which the partial atomic charges in the classical model are scaled (most often by a factor of 0.8 by setting the total charge of the ions to ± 0.8) can be applied. Through this approach, not only reproducing the experimental dynamic properties becomes available, but it has also been shown to relate to the physical effect of charge transfer between cation and anion.⁵⁴ Despite the simplicity of charge scaling, it is not self-evident what scaling factor one should take for the particular liquid, especially in a mixture of ionic and neutral substances, as is the case for a DES. Furthermore, it has been argued that this approach would require the complete refitting of nonbonding parameters within the force field, since otherwise scaling may induce significant errors in the structure of the hydrogen bonding interactions in analogous IL systems.⁵⁰ Following from these considerations, systems were simulated twice in this study: once with no charge scaling and once with scaling by a factor of 0.8 on charges for the glycine component ions choline and chloride. By scaling down the charges, the diffusion of the component ions indeed becomes faster. However, the hydrogen bonding structure of the liquid changes severely (refer to Figure 6 and Table S1 in the Supporting Information), indicating that charge scaling introduces errors to the MD simulations. For this reason, only the MD

simulations on systems with no charge scaling will be further discussed in this article.

The initial cubic simulation box was prepared by the PACKMOL⁵⁵ program from a microfibril of beta-cellulose (composed of 10 glucose octamer chains), 500 ion pairs of choline chloride, and 1000 molecules of glycerol (system I). A second simulation box was also created from the pure solvent, using 500 ion pairs of choline chloride and 1000 molecules of glycerol (system II) as a reference. After an initial energy minimization performed on these boxes under periodic boundary conditions to remove the possible hotspots in the structure, the systems were simulated in an isothermal–isobaric (NPT) ensemble for 1 ns under 1 bar pressure and 300 K temperature. Through the last 0.5 ns, the volume of each box was averaged and the obtained average volumes were used for the subsequent runs. After 1 ns of NVT ($T = 300$ K) equilibration, a 50 ns long run was performed, during which the atomic coordinates were saved every 1 ps. The structure of the system was analyzed by the TRAVIS⁵⁶ program. Surface coverage data were obtained by using Voronoi analysis, as implemented in TRAVIS.⁵⁷

3. RESULTS AND DISCUSSION

3.1. Bacterial Cellulose Eutectogel Preparation. Glyceline-containing bacterial cellulose eutectogels (Gly-BCEGs) were prepared according to the experimental procedures described in the Experimental Section above. Notably, when gels containing more than 99 wt % DES were dried, they displayed more hygroscopicity even than neat glyceline. As a result, we found it necessary to heat the gels (~ 60 °C) in order to reduce water contents to levels comparable to that of neat glyceline. After drying both neat glyceline and glyceline eutectogels, residual water persisted, as is evident in the respective no-D ^1H NMR spectra shown in Figure 1 (residual water appears near 4.25 ppm). Based on previous reports of the effects of water on diffusion rates in neat glyceline,³¹ the water contents were adjusted using NMR signals as a guide in order to achieve approximately the same water content in

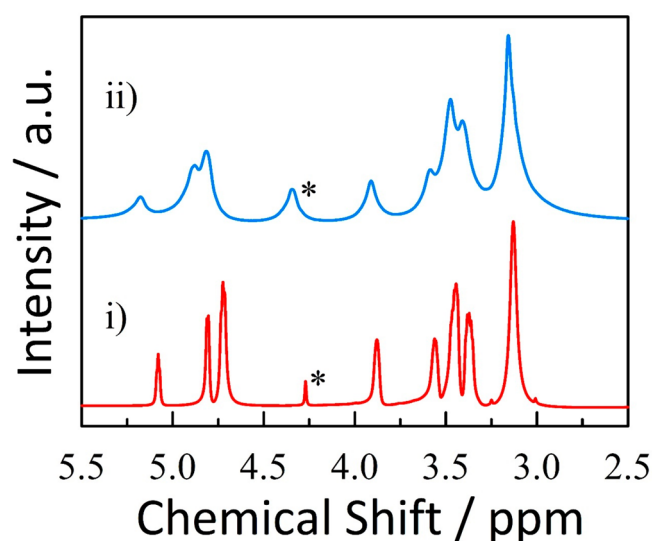


Figure 1. No-D ^1H NMR spectra collected for (i) neat glyceline and (ii) a BCEG containing 98.8 wt % glyceline. The * denotes the presence of a similar amount of residual water in both free glyceline and the corresponding glyceline-based eutectogel.

order to equitably examine glyceline dynamics in the free and confined states. We note that, although a plausible route to even lower water contents, further drying of these eutectogels under vacuum was not examined.

3.2. Small-Angle Neutron Scattering. Deuterated BCEGs hosting 50:50 and 99:1 mass ratios of D-Gly/D₂O were analyzed by SANS (see Figure 2). These BCEGs,

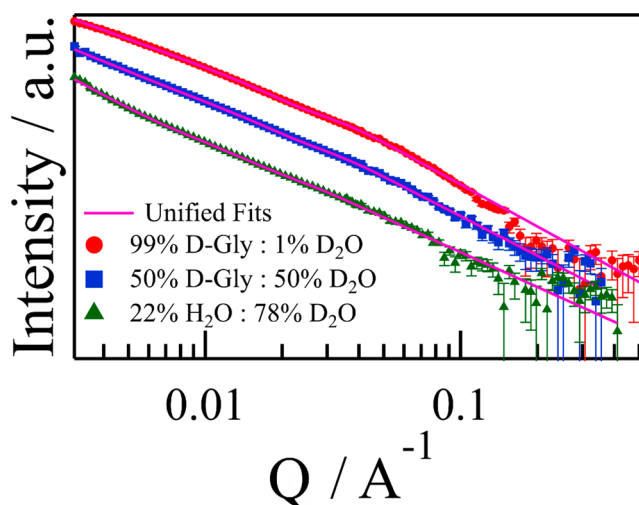


Figure 2. SANS profiles measured for bacterial cellulose containing three different solvent mixtures: (i) D-Gly/D₂O at 99%:1% (red circles), (ii) D-Gly/D₂O at 50%:50% (blue squares), and (iii) H₂O/D₂O at 22%:78% (green triangles) and the corresponding Unified Fit analysis of the SANS data. The indicated ratios denote mass ratios for the incorporated solvent phase. The 50%:50% and 99%:1% D-Gly/D₂O profiles have been scaled by factors of 10 and 100, respectively, to offset the data for the purpose of clarity.

containing vastly different amounts of D₂O, served to simulate the effects of the dehydration process on BCEGs. A bacterial cellulose hydrogel containing H₂O and D₂O in a 22:78 mass ratio was employed as a control. This particular H₂O/D₂O ratio within the hydrogel was selected to provide a reasonable contrast to the completely hydrogenated cellulose fibers. The SANS data were recorded for a Q range of $0.003\text{--}0.8\text{ Å}^{-1}$ and analyzed using the Unified Fit Approach (UFA).³⁸ This fitting method provides valuable insights into structural features, such as the radius of gyration (R_g) and extent of entanglement (P) of the cellulose fibers. We note that UFA was specifically developed for the analysis of hierarchical systems and has been successfully applied to the study of biomaterials similar to bacterial cellulose, including lignocellulose, hemicellulose, cellulose, and switchgrass.^{58–60} The SANS data for BCEG samples containing D-Gly and D₂O were fitted to eq 1 by using level-1 and level-2 structural parameters of the UFA (see section 2.6 of the Experimental Section for details), which covered the Q -ranges $0.003\text{--}0.02\text{ Å}^{-1}$ (level-2) and $0.009\text{--}0.7\text{ Å}^{-1}$ (level-1) with an overlap Q -region of $0.009\text{--}0.02\text{ Å}^{-1}$ for structural interpretation. However, the SANS data from the control hydrogel were fitted to eq 1 using only level-1 parameters for structural interpretation. The structural parameters recovered using the Unified Fit are summarized in Table 1.

The UFA analysis revealed an R_g of $253 \pm 31\text{ Å}$ for the cellulose microfibril in the hydrogel which is similar to the previously reported R_g value for the cross-section of the cellulose microfibril for BC.⁵⁹ Interestingly, compared to the

Table 1. Unified Fit Parameters to SANS Data Acquired for Deuterated Glyceline at Two Hydration Extremes Confined in Bacterial Cellulose and for a Control Bacterial Cellulose Hydrogel

liquid in gel	1st level				2nd level	
	Guinier		power-law		power-law	
	<i>G</i>	<i>R_g</i> (Å)	<i>B</i> (× 10 ^{−6})	<i>P</i>	<i>B</i> (× 10 ^{−5})	<i>P</i>
D-Gly/D ₂ O						
99:1	2.8 ± 0.2	109 ± 15	6.3 ± 2.3	3.3 ± 0.1	27 ± 7	2.2 ± 0.1
50:50	2.4 ± 0.1	117 ± 24	7 ± 3	3.1 ± 0.1	5.0 ± 1.2	2.4 ± 0.1
H ₂ O/D ₂ O						
22:78	9.0 ± 0.2	253 ± 31	14 ± 3	2.6 ± 0.1		

hydrogel, the R_g values for the cellulose microfibril decreased significantly to 109 ± 15 Å and 117 ± 24 Å in BCEGs containing 1% and 50% D₂O, respectively. This suggests a collapse in the microfibrillar structure of the BC that may occur as a result of removal of water molecules from the interfibrillar space. This dehydration of the interfibrillar space of cellulose can be attributed to the higher affinity of the DES for interactions with water via hydrogen bonding compared with its interactions with cellulose. This collapse of the cellulose microfibril is further supported by the level-1 power-law (*P*) values that describe the extent of entanglement of these microfibrils. The *P* value for the control hydrogel is 2.6, which is indicative of a highly entangled network of hydrated cellulose microfibrils that form a larger macrofibril. In contrast, the *P* values of the BCEG samples containing 1% and 50% D₂O are 3.3 ± 0.1 Å and 3.1 ± 0.1 Å, respectively, indicating a collapse of the microfibrillar structure and a reduction in entanglement of the microfibrils. The change in the level-1 *P* behavior from a mass-fractal ($2 < P < 3$) in the hydrogel to a surface fractal ($3 < P < 4$) in the BCEGs is consistent with the proposed change in the microfibrillar structure from a hydrated network of cellulose microfibrils to a coalesced structure with rough interfaces, a phenomenon that stems from the removal of water molecules from the interfibrillar space between the microfibrils.

The level-2 structural regime ($Q = 0.003\text{--}0.02$ Å^{−1}) provides information about the arrangement of the microfibrils within the bacterial cellulose network. The level-2 *P* values were obtained by fitting the data in eq 1 for the Gly-BCEGs containing 50:50 and 99:1 mass ratios of D-Gly/D₂O. The results indicate that the cellulose microfibrils in these samples are loosely entangled, forming a sparsely packed network. This suggests that the vast majority of the solvent is contained by large micropores within the cellulosic structure, implying mesoscopic confinement within bacterial cellulose gels, as appears to be the case for [emim][Tf₂N] studied previously.³⁴

3.3. X-ray Diffraction. XRD patterns were measured for Gly-BCEGs hosting varying amounts of glyceline from 18.4 to 94.8 wt % (Figure 3). The bacterial cellulose crystal structure gives rise to a [200] peak at 22.7°, a [110] peak at 16.8°, and a [110] peak at 14.4°. These XRD data are well described by multiple, cumulative Voigt profiles, yielding correlation coefficient (r^2) values in the 0.9938–0.9968 range. These fit values are similar in numerical quality to previously published XRD studies of bacterial cellulose ionogels,³² indicating the generality of Voigt peak shapes for adequately describing complex XRD profiles. XRD fitting analysis conducted on Gly-

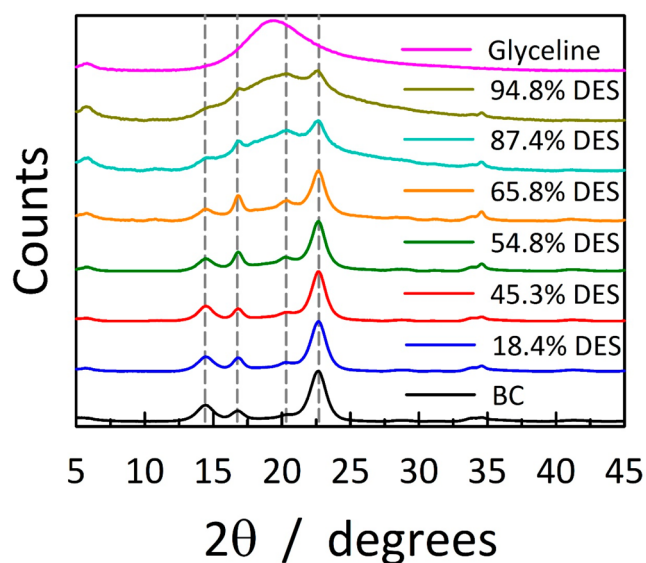


Figure 3. XRD patterns measured for BC, BCEGs containing varying contents of glyceline (the indicated percentages of 18.4–94.8% are based on mass fraction) and neat glyceline. Dashed vertical gray lines represent the positions of crystalline BC peaks.

BCEGs for increasing contents of glyceline reveals that the incorporation of glyceline does not perturb or diminish the crystallinity of the host bacterial cellulose (Table 2). Similar observations were made for bacterial cellulose ionogels earlier.³² Crystallite sizes (τ) were determined from peak fitting results using the Scherrer equation (eq 2); these results are also captured in Table 2. Monitoring the [110] reflection, the crystallite size slowly increases with glyceline content from 18.4 to 87.4 wt % before then falling substantially (nearly 2-fold) at 94.8 wt % glyceline. Similarly, the [200]-derived crystallite size holds relatively constant at ~ 76 Å across the range of 18.4–87.4 wt % glyceline before dropping slightly to 66 Å. Finally, for the [110] reflection, the crystallite size is static at ~ 67 Å for 18.4–65.8 wt % glyceline but increases to 75 Å at the highest glyceline loading of 94.8 wt %. In contrast to the current eutectogel results, the bacterial cellulose crystallite size determined in ionogels based on the [200] and the [110] reflections varied little with ionic liquid content but showed a monotonic size increase based on the [110] reflection.³² We tentatively attribute these observed changes to the cellulose crystal domains within BCEGs to the removal of water molecules from the interfibrillar space between cellulose microfibrils, mediated by the glyceline DES, resulting in collapse and coalescence of the cellulose microfibrils in the BCEGs.⁶¹

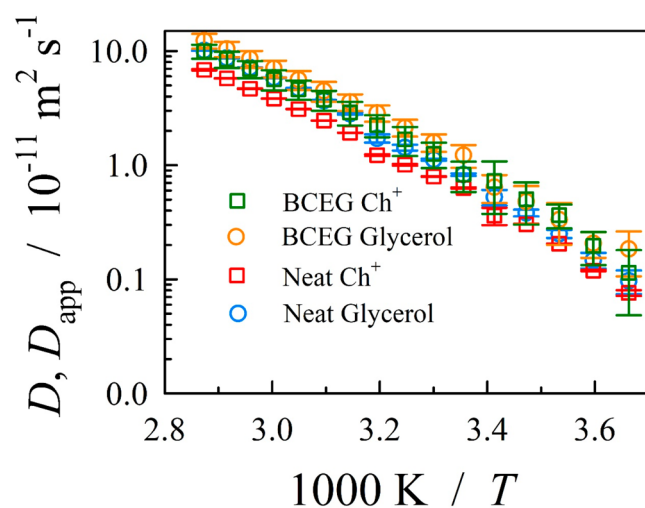
3.4. Dynamic NMR. A comparison between the ¹H NMR spectrum of neat glyceline (Figure 1i) and a typical Gly-BCEG (Figure 1ii) reveals clear changes in the hydrogen bonding network upon confinement of glyceline in bacterial cellulose. The chemical shifts of the glyceline protons in Gly-BCEG appear downfield of those for free glyceline, indicative of deshielding in the eutectogel. For example, the methyl peak for choline is shifted downfield from 3.13 to 3.28 ppm upon glyceline incorporation within the BCEG. Similar shifts are evident for the hydroxyl protons of glycerol and choline upon eutectogel incorporation. That is, the choline (5.08 ppm) and glycerol (4.81 and 4.73 ppm) hydroxyl peaks in free glyceline shift downfield to 5.18, 4.88, and 4.82 ppm, respectively, upon

Table 2. Experimental and Calculated XRD Parameters for a Bacterial Cellulose Aerogel and Glyceline-Containing Bacterial Cellulose Eutectogels

glyceline	CrI	reflection [200]			[110]			[1 $\bar{1}$ 0]			r^2
		$2\theta_{\max}$ (deg)	β	τ (Å)	$2\theta_{\max}$ (deg)	β	τ (Å)	$2\theta_{\max}$ (deg)	β	τ (Å)	
BC	0.88	22.7	1.34	67	16.8	1.13	79	14.4	1.46	61	0.9995
18.4	0.88	22.7	1.18	76	16.8	0.81	110	14.4	1.32	67	0.9938
45.3	0.88	22.7	1.19	76	16.8	0.79	113	14.4	1.31	68	0.9968
54.8	0.85	22.7	1.15	78	16.8	0.77	116	14.4	1.30	68	0.9949
65.8	0.86	22.7	1.20	75	16.8	0.69	129	14.4	1.35	66	0.9954
87.4	0.84	22.7	1.16	78	16.8	0.67	133	14.4	1.22	73	0.9938
94.8	0.90	22.7	1.37	66	16.8	1.30	69	14.4	1.19	75	0.9966

mesoscopic confinement within the BCEG, a feature pointing to a perturbation in the intermolecular hydrogen bonding of glycerol and the choline cation induced by the cellulose surface. The chemical shift assignments for neat glyceline and a representative Gly-BCEG containing 98.8 wt % glyceline are summarized in the [Supporting Information](#).

Proton DOSY NMR experiments were conducted on neat glyceline and Gly-BCEGs in order to understand the impact of mesoscopic confinement on the mobility of the individual components of the DES upon incarceration within bacterial cellulose. The DOSY NMR results ([Figure 4](#) and [Table S2](#))

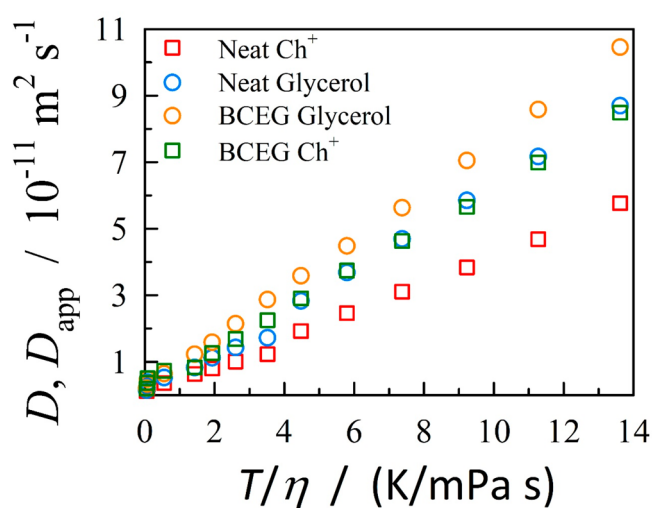
**Figure 4.** Temperature dependence (258–348 K) of the molecular diffusion coefficients (D) and apparent diffusion coefficients (D_{app}) for choline and glycerol components in neat glyceline and glyceline BCEGs. The glyceline loading in the BCEG is 98.8 wt %.

clearly demonstrate faster long-range translational diffusion rates for glycerol compared to choline in free glyceline, an observation in accord with previously reported diffusional studies of glyceline using quasi-elastic neutron scattering and pulse-field gradient NMR techniques.^{25,30,62}

Surprisingly, however, and in contrast to results for ionogel-confined ILs,^{34,63} mesoscopic confinement of glyceline within a bacterial cellulose eutectogel produced a notable increase in the apparent diffusion (D_{app}) rate of the component species. That is, while glycerol diffused faster than choline, both species exhibited diffusion rates in the eutectogel that were significantly faster than for the free DES. The origin of this behavior is not immediately clear. We hypothesize that the increased diffusion rates for glycerol and choline in the BCEG arise from partial migration of chloride anion from the bulk to

the cellulose microfibril surface where it interacts with the hydroxyl groups of the sugar units in the cellulose fibers which act as competitive hydrogen bond donors. Indeed, chloride is a well-known hydrogen bond acceptor and plays a central role in choline chloride-based DES liquid structure via interactions with hydrogen bond donors (such as glycerol) to form transient cage-like structures which retard diffusion. Reduction of a fraction of the activity of chloride by surface confinement may reduce interactions with glycerol and choline, resulting in the observation of enhanced diffusion rates. Moreover, due to its negative charge, chloride is a stronger hydrogen bond acceptor than water. As a result, strong interactions between chloride and sugar hydroxyl units of the bacterial cellulose backbone likely contribute to the displacement of water molecules from the interfibrillar space, resulting in the collapse of the microfibrillar structure revealed by SANS. To test this hypothesis of preferential cellulose surface interactions with chloride leading to a “plasticization” of the DES fluidity under mesoscopic confinement, it is essential to gain a deeper understanding of the intermolecular interactions taking place between glyceline components (i.e., choline, chloride, glycerol) when presented with a cellulose surface.

The diffusion data from [Figure 4](#) are presented as Stokes–Einstein plots in [Figure 5](#) in order to further examine the diffusive behavior of glyceline components before and after

**Figure 5.** Plots of the diffusion coefficients D and D_{app} against T/η , demonstrating the conformity of the data to Stokes–Einstein behavior. Diffusion coefficient data from [Figure 4](#) were used to generate these plots. Note that the temperature-dependent viscosities for glyceline BCEGs in this figure were assumed to conform to the neat glyceline values.

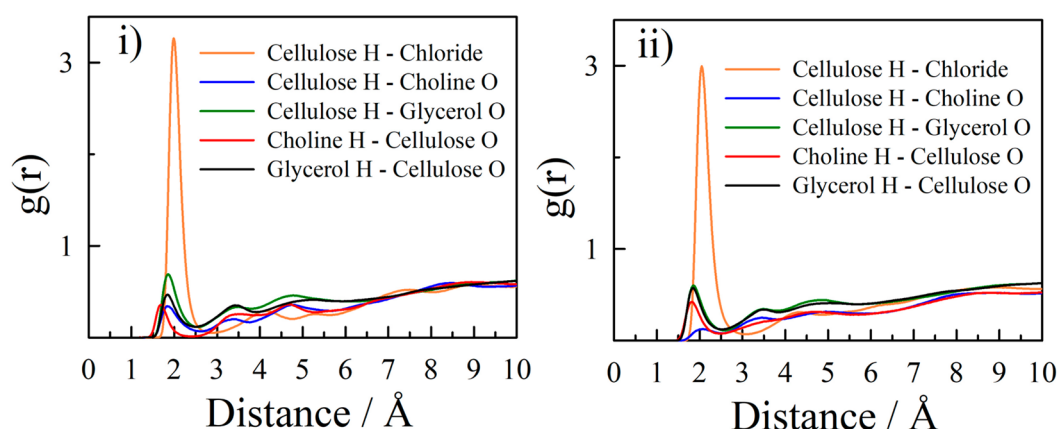


Figure 6. Radial distribution functions (RDFs) representing the individual cellulose–solvent interactions in the MD simulations with unity charges on the ions (i) and with scaled charges (ii).

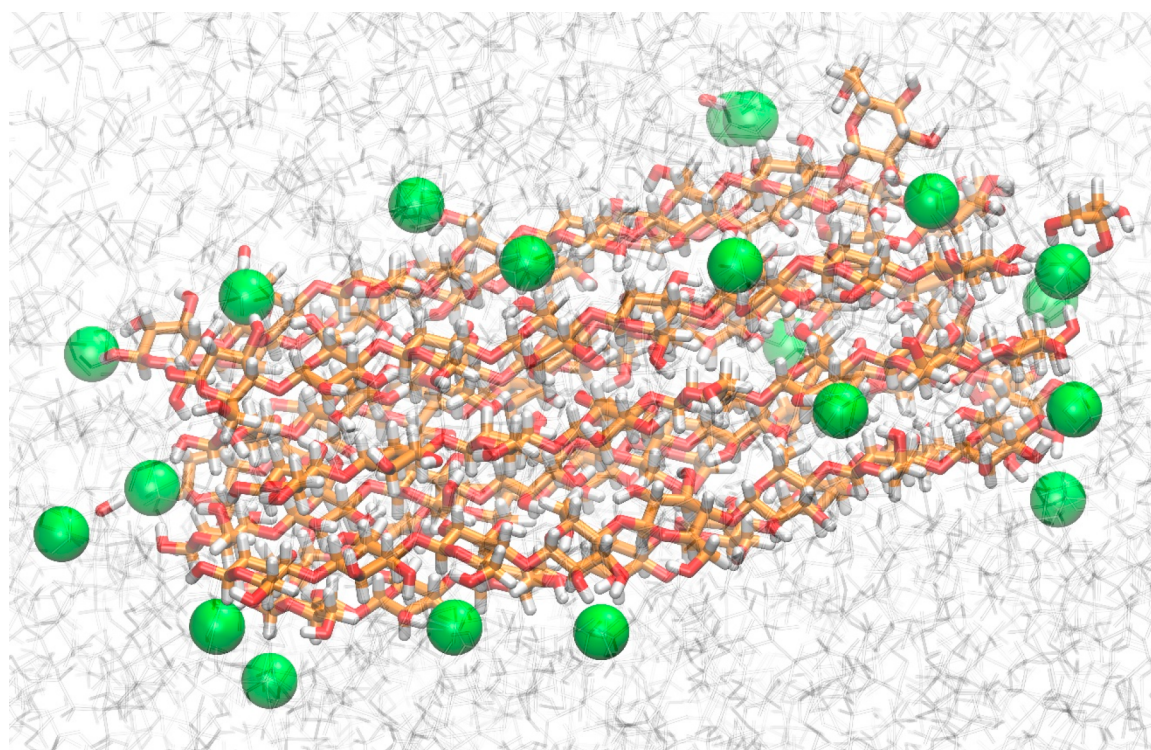


Figure 7. Schematic showing a van der Waals depiction of chloride (green) at the surface of cellulose captured by MD simulations through Voronoi analysis.

confinement. The medium viscosity (η) within the eutectogel is inaccessible to conventional measurements (e.g., rheology); therefore, the assumption was made that the viscosity in the gel is the same as the viscosity of the bulk liquid (glyceline) at a given temperature. As a consequence, the Stokes–Einstein plots can only be used to elucidate general trends and resist a more detailed analysis. In spite of this, it is evident that, in both cases (free and confined glyceline), the data conform to the Stokes–Einstein relation with no obvious departure. Certainly, no “slip” or “stick” behavior can be evoked from these data. Presented in this manner, the data also resonate with the earlier observations that (i) contrary to prediction, mesoscopic confinement within the eutectogel enhances the diffusion of both species and (ii) glycerol diffuses faster than choline, whether glyceline is free or confined. We note that this latter observation follows the notion that choline tends to move

faster over shorter distances,²⁵ while the glycerol is faster in terms of long-range, translational diffusion. Closer inspection of the graph also reveals that (iii) the degree of diffusional enhancement upon confinement is larger for choline. A possible explanation for this behavior is that choline experiences a higher relative increase in diffusion rate upon mesoscopic confinement because it is associated with a single hydrogen bond donor (hydroxyl) whereas glycerol presents three hydroxyl groups. Following this logic, it is possible that a reduction in chloride activity resulting from molecular entanglement (hydrogen bonding) with cellulose microfibrils liberates the choline to a greater extent than glycerol, although both clearly show increased mobility. If the Stokes–Einstein relation is valid, then the quantity $D\eta/T$ should be constant. This implies that the effective hydrodynamic radius must remain constant for choline and glycerol in a given system (i.e.,

free or BCEG-confined), confinement decreases the effective hydrodynamic radius of both species, and the decrease in the effective hydrodynamic radius is greater for choline than for glycerol. Illuminating these interfacial interactions is nontrivial. In order to glean key molecular insights, we turn our attention to MD simulations.

3.5. Molecular Dynamics Simulations. The above DOSY NMR experiments reveal the diffusion dynamics of choline and glycerol within free glycine and BCEG-confined glycine but lack the ability to probe the pivotal role of chloride in these systems. To uncover the role of chloride, we have performed MD simulations on a single cellulose microfibril which serves as a representative beta-cellulose model to probe the key molecular interactions between glycine components and bacterial cellulose. The microfibril model, comprising 10 glucose octamer chains, was applied previously to interrogate interactions between cellulose and ions of an ionic liquid present within an ionogel.³⁴

The MD simulations reveal no significant changes in the cellulose structural features, including the distance between individual cellulose chains in the strand and torsion angles between the sugar units in the cellulose chains, consistent with the retention of crystallinity demonstrated in XRD measurements performed on BCEGs. This agreement between MD simulations and XRD experiments, coupled with the proven ability of MD simulations to reproduce the dissolution of cellulose in certain ILs using this cellulose model, advocates for the suitability of the selected cellulose model.^{64,65}

A bacterial cellulose eutectogel is a spongy material accommodating DES in the entwined regions of cellulose fibrils which introduce a significant surface area for interactions with DES. Insight into the molecular interactions prevalent at this cellulose–DES interface are thus essential to understanding the behavior of a BCEG. Among the many possible interactions between the glycine components and a model cellulose microfibril, hydrogen bonding between chloride and the cellulose surface is overwhelmingly dominant, as illustrated in the radial distribution functions (RDFs) presented in Figure 6. This observation adds insight to the results obtained from diffusional NMR, in particular, the increased apparent diffusion rates for choline and glycerol upon eutectogel confinement. That is, the MD simulation results add support to the notion that the faster diffusion rates stem from migration of chloride from the bulk to the cellulose microfibrillar surface (Figure 7). This is consistent with our earlier proposal that preferential cellulose interactions with chloride lead to a “plasticization” of the glycine fluidity under mesoscopic confinement within the eutectogel. A reduction in the chloride concentration in the interfibrillar space likely manifests in a reduced number of hydrogen bonding interactions between chloride and its usual partners (choline and glycerol). This decoupling is accompanied by a smaller effective hydrodynamic radius for these species which is reflected in higher diffusion rates. Chloride migration to the cellulose microfibril modulates the electronic environment surrounding choline and glycerol, explaining the deshielding of the proton peaks observed for these components in the NMR spectrum upon Gly-BCEG confinement. In addition, the proposed sorption of chloride on cellulose microfibrils is expected to perturb the nature of the hydrogen bonding on the cellulose surface by displacing water molecules, leading to dehydration of cellulose and the eventual collapse of the microfibrillar structure suggested in complementary SANS analyses.

Apart from the predominant chloride–cellulose interactions, glycerol–cellulose hydrogen bonding represents the next strongest interaction followed by choline–cellulose interactions (Figure 6). This order in interaction strengths results in a specific structure at the cellulose–glyceline interface, as shown in Figure 8 [panel (ii)] is the interaction surface of a model

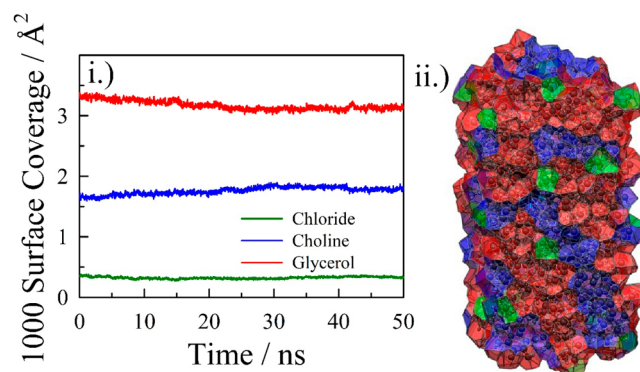


Figure 8. (i) Time development of the surface coverage of a cellulose microfibril in glycine by the three components of glycine, calculated by Voronoi analysis. (ii) The interaction surfaces of cellulose in a snapshot of the simulation is also represented, indicating the color-coded location of the chloride (green), choline (blue), and glycerol (red) components at the interface. The image was created by showing and coloring those faces of the Voronoi cells in the system which are shared between the cellulose and the solvent components and thus belong to the interface.

cellulose fibril generated by Voronoi analysis of an MD simulation where components of the glycine at the interface are color-coded green (chloride), blue (choline), and red (glycerol)]. Apparently, considering that it is the smallest particle in the system, the chloride anion occupies a remarkable area on the carbohydrate surface, in agreement with NMR experiments and the RDFs. While choline is also adsorbed on the cellulose surface to a considerable extent, as shown by the fairly large surface coverage by this species, its relatively weaker hydrogen bonding with the microfibril suggests that its interfacial accumulation is mainly governed by its electrostatic attraction to chloride anions adsorbed on the surface. Indeed, choline and chloride ions generally occupy adjacent regions on the cellulose surface, as shown in Figure 8(ii). The largest fraction of the surface is, however, covered by glycerol molecules which can form multiple hydrogen bonds with the cellulose surface. The spatial distribution of the different species at the surface appears to suggest aggregation of glycerol and charged species into separate domains, analogous to the microheterogeneity in liquid structure suggested for ILs and DESs.^{27,66–68} Through interactions with the neutral glycerol as well as the ionic components of glycine, the cellulose is apparently incorporated into both mesoscale domains of the liquid.

4. CONCLUSIONS

In this work, we introduce bacterial cellulose eutectogels incorporating glycine at extremely high loadings (up to 98.8% by weight) and explore the behavior of the entrapped fluid as well as the cellulosic host by a suite of experimental means. X-ray diffraction analyses show that the naturally high crystallinity of the bacterial cellulose is virtually unaffected by the resident glycine contained within the gels. These

observations are consistent with molecular dynamics simulations which indicate no structural changes for a model beta-cellulose microfibril solvated by glyceline. Small-angle neutron scattering experiments reveal a significant decrease in the radius of gyration for bacterial cellulose within glyceline-laden eutectogels, attributed to a contraction of the microfibril structure resulting from loss of water molecules. Proton NMR peaks for glyceline in a eutectogel appear downfield (deshielded) with respect to peaks for bulk glyceline. Diffusion NMR experiments establish faster apparent diffusion rates for glycerol and choline mesoscopically confined in a eutectogel than for these components in neat glyceline. MD simulations provide an important framework for understanding these results and point to the migration of chloride from the bulk to the cellulose surface as the origin of this anomalous diffusion behavior. The consequences of the chloride mobilization resulting from cellulose–glyceline interactions include modulation in the nature of the hydrogen bonding, both interfacially and in the interfibrillar space, and dehydration and collapse of the microfibrillar structure. Finally, we note that, although they provided critical insights which helped to shed light on startling experimental outcomes, classical MD simulations are unable to reproduce the experimentally observed increase in the diffusion constants of the glyceline components in the presence of cellulose, pointing to the urgency of improved force fields (e.g., polarizable models) to accurately describe the structure and the dynamics within deep eutectic solvent-based systems, such as eutectogels.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpcb.0c04916>.

Supplemental figures of XRD peak-fitting analysis, NMR assignments, and MD simulations showing radial distribution functions plus additional details on MD simulations (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Gary A. Baker – Department of Chemistry, University of Missouri-Columbia, Columbia, Missouri 65211, United States; orcid.org/0000-0002-3052-7730; Phone: 573-882-1811; Email: bakergar@missouri.edu

Authors

Chip J. Smith II – Department of Chemistry, University of Missouri-Columbia, Columbia, Missouri 65211, United States; orcid.org/0000-0003-3621-9429

Durgesh V. Wagle – Department of Chemistry and Physics, Florida Gulf Coast University, Fort Myers, Florida 33965, United States; orcid.org/0000-0002-2522-0670

Nakara Bhawawet – Department of Chemistry, Faculty of Science, Chulalongkorn University, Bangkok 10330, Thailand; orcid.org/0000-0001-7381-4847

Sascha Gehrke – Mulliken Center for Theoretical Chemistry, University of Bonn, Bonn 53115, Germany

Oldamur Hollóczki – Mulliken Center for Theoretical Chemistry, University of Bonn, Bonn 53115, Germany; orcid.org/0000-0003-3105-7176

Sai Venkatesh Pingali – Biology and Soft Matter Division, Oak Ridge National Laboratory, Oak Ridge, Tennessee 37831, United States; orcid.org/0000-0001-7961-4176

Hugh O'Neill – Biology and Soft Matter Division, Oak Ridge National Laboratory, Oak Ridge, Tennessee 37831, United States; orcid.org/0000-0003-2966-5527

Complete contact information is available at:

<https://pubs.acs.org/doi/10.1021/acs.jpcb.0c04916>

Notes

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